MemDesAI 2027

International Conference on Membrane Technology, Desalination, Artificial Intelligence and Environmental Sustainability

6 – 8 December 2027

IIT Roorkee, Uttarakhand, India

Jointly organized by

Indian Institute of Technology Roorkee

CSIR - Indian Institute of Petroleum, Dehradun

In collaboration with

Department of Chemical and Biological Engineering, Hong Kong University of Science & Technology (HKUST)

250–300 Delegates

3 Days Plenary, tracks & posters

30–40 Invited Speakers

4 Technical Tracks

Outcomes:

- Peer-reviewed proceedings
- Special issue in reputed journal
- MoUs with international partners
- Best paper/poster awards
- Networking & collaboration
- Technology demonstration
- Brainstorm sessions (Academic-Industry)
- Hands-on sessions
- Scientific luminaries honour

Convener:
Dr. Bhaskar Jyoti Deka, IIT Roorkee

Co-Conveners:
Dr. Racha Surya Murali, CSIR-IIP
Dr. Anshul Yadav, IIT Roorkee

Save the date — abstract submission and registration details to follow

Why this conference matters

Membrane separation is one of the most energy-efficient platforms for water purification and resource recovery. MemDesAI-2027 brings together academia, national laboratories and industry to align three accelerating trends — affordable low-carbon desalination, AI-driven design and process control, and environmentally sustainable membrane manufacturing — on a single, structured platform for India and its global partners.

Conference Themes

Membrane Technology

Nanocomposite & TFC membranes, fabrication and characterization

Role of AI in Membranes

AI assisted membrane fabrication, ML-assisted design, fouling prediction, digital twins & process control

Desalination

Seawater & brackish treatment, brine management, resource recovery

Water & Wastewater Treatment

Contaminant removal and process-scale treatment systems

Air Purification

Membrane-based air filtration and pollutant capture

Gas Separation

CO_2 capture and industrial gas separation membranes

Geomembranes

Containment, lining systems and civil/environmental applications

Fuel Cells

Membrane materials for energy conversion & storage systems

Biomedical & healthcare

Hemodialysis, drug delivery, blood purification, tissue engineering, biosensors, and diagnostics

CSIR - IIP

CBE - HKUST

IIT Roorkee